

PHYSICS FINAL EXAM GUIDE 2026

Strategic High-Yield Study Manual (Sets 55/7/1, 55/7/2, 55/5/2, SR5QP/5)



THE "SURE-SHOT" 5-MARKERS

- 1. Optical Instruments:** Ray diagram for **Compound Microscope / Astronomical Telescope** (Normal Adjustment). Derivation of Magnifying Power (m).
- 2. Semiconductors:** Explain **Full Wave Rectifier**. Requirements: Circuit Diagram, Working logic, Input/Output Waveforms.
- 3. Electrostatics:** **Parallel Plate Capacitor** with a dielectric slab of thickness ' t '. Analyze factors (C, Q, V, E, U) when battery is connected vs. disconnected.
- 4. Magnetism:** **Moving Coil Galvanometer** Construction & Working. Explain Radial Magnetic Field and conversion to Ammeter/Voltmeter.



ESSENTIAL DERIVATIONS (Checklist)

- ☐ **Huygens' Principle:** Verification of Laws of Refraction (Snell's Law) or Reflection.
- ☐ **Bohr's Model:** Derivation of the Radius of n -th orbit and Total Energy.
- ☐ **Drift Velocity:** Expression in terms of relaxation time (τ) and Electric field (E).
- ☐ **Current-Velocity:** Establish relation $I = nAev_d$.
- ☐ **Lens Maker's Formula:** Standard focal length derivation for lenses.



RECURRING GRAPHS & CONCEPTS

Photoelectric Effect: Graph of **Stopping Potential (V_0) vs Frequency (ν)**. Slope = h/e , X-intercept = Threshold frequency, Y-intercept = Work function.

Nuclear Physics: The fact that **Nuclear Density is Constant**. Result of density ratio questions is always 1:1.

AC Circuits: Resonance in series LCR circuit. Condition: $X_L = X_C$ or $\omega = 1/\sqrt{LC}$.



FINAL PREPARATION CHECKLIST

- ☐ Ray Diagrams: Verify arrow directions on every light ray.

- ☐ Waveforms: Sync input and output peaks for rectifiers.

- ☐ Units & Dimensions: Confirm Planck's constant units and work function in eV.

- ☐ Calculator Check: For numericals, check the power of 10 in Bohr's constants.
